12TH BIOLOGY · CHAPTER 13 · FULL CHAPTER MCQS

Thermoregulation and Osmoregulation

70 topic-wise MCQs from the complete chapter with answers and reasons

13.1 HOMEOSTASIS

1. The maintenance of a stable internal environment despite changes in the external environment is called:

(a) Homeostasis (b) Excretion
(c) Metabolism (d) Osmosis

Answer: (a) Homeostasis — This is the definition of homeostasis.

2. Which of these is NOT listed as an important process of homeostasis?

(a) Thermoregulation (b) Osmoregulation
(c) Digestion (d) Excretion

Answer: (c) Digestion — The important processes are thermoregulation, osmoregulation and excretion.

3. The structures that detect changes in the internal or external environment are:

(a) Effectors (b) Receptors
(c) Control centres (d) Set points

Answer: (b) Receptors — Receptors (sensors) detect changes and inform the control centre.

4. Which of these act as control centres in homeostasis?

(a) Muscles and glands
(b) Kidneys and liver
(c) Skin and blood vessels
(d) Brain, spinal cord and endocrine glands

Answer: (d) Brain, spinal cord and endocrine glands — They receive information, process it and send messages to effectors.

5. Which of these act as effectors?

(a) Brain and spinal cord (b) Muscles and glands
(c) Receptor cells of skin (d) Thermoreceptors

Answer: (b) Muscles and glands — Muscles respond by contracting and glands by releasing secretions.

6. The optimal values at which the internal environment is maintained are called:

(a) Set points (b) Stimuli
(c) Feedbacks (d) Responses

Answer: (a) Set points — Set points are optimal for the body's function.

7. Changes in blood pressure are detected by receptor cells present in the:

(a) Skin (b) Muscles
(c) Blood vessels (d) Liver

Answer: (c) Blood vessels — Receptor cells in blood vessels detect changes in blood pressure.

8. When body temperature drops below the set point, effectors in the skin and muscles cause:

(a) Sweating and vasodilation (b) Vasodilation only
(c) Sweating and shivering (d) Shivering and vasoconstriction

Answer: (d) Shivering and vasoconstriction — These responses increase heat production and reduce heat loss.

9. The correct sequence in a homeostatic mechanism is:

(a) Effector → receptor → control centre
(b) Receptor → control centre → effector
(c) Receptor → effector → control centre
(d) Control centre → receptor → effector

Answer: (b) Receptor → control centre → effector — Receptors detect, the control centre processes, and effectors respond.

10. A mechanism in which the output of a system reduces the activity of the system is:

(a) Positive feedback (b) Osmoregulation
(c) Negative feedback (d) Vasodilation

Answer: (c) Negative feedback — This is negative feedback, the most common form of regulation.

11. Which of these is an example of negative feedback?

(a) Blood clotting (b) Childbirth
(c) Labour (d) Regulation of blood glucose level

Answer: (d) Regulation of blood glucose level — Body temperature, blood glucose and hormone secretion are regulated by negative feedback.

12. Which of these is an example of positive feedback?

(a) Blood clotting (b) Regulation of blood glucose
(c) Regulation of body temperature (d) Hormone secretion

Answer: (a) Blood clotting — Positive feedback occurs in blood clotting, labour and childbirth.

13. Which hormone is released by the pituitary gland during childbirth?

(a) Thyroxine (b) Oxytocin
(c) Insulin (d) Cytokine

Answer: (b) Oxytocin — Oxytocin stimulates further contractions of the uterus.

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14. The positive feedback loop of childbirth continues until:

(a) Oxytocin is finished
(b) The uterus relaxes
(c) The baby is born
(d) Body temperature falls

Answer: (c) The baby is born — The loop stops when the baby is born.

15. Positive feedback:

(a) Reverses the original stimulus
(b) Always stops the stimulus
(c) Has no effect on the stimulus
(d) Enhances the original stimulus

Answer: (d) Enhances the original stimulus — The output increases the activity of the system.

16. The proteins released by immune cells in severe inflammation or infection are:

(a) Cytokines
(b) Keratins
(c) Chitins
(d) Purines

Answer: (a) Cytokines — Cytokines further stimulate the immune response.

17. Unregulated positive feedback in severe infection can lead to:

(a) Gout
(b) Excessive inflammation and tissue damage
(c) Hypothermia
(d) Dehydration

Answer: (b) Excessive inflammation and tissue damage — Uncontrolled cytokine release damages tissues.

18. The set point of human body temperature is:

(a) 25 °C
(b) 98 °C
(c) 40 °C
(d) 37 °C

Answer: (d) 37 °C — Humans maintain their temperature at about 37 °C.

13.2 OSMOREGULATION

19. The maintenance of the balance of water and solutes in body fluids is called:

(a) Thermoregulation
(b) Excretion
(c) Osmoregulation
(d) Anhydrobiosis

Answer: (c) Osmoregulation — This is the definition of osmoregulation.

20. Animals whose body fluids remain isotonic to their environment are:

(a) Osmoconformers
(b) Osmoregulators
(c) Homeotherms
(d) Endotherms

Answer: (a) Osmoconformers — Osmoconformers let their body fluids match the environment.

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21. Which of these is an osmoconformer?

(a) Kangaroo rat (b) Shark
(c) Freshwater fish (d) Human

Answer: (b) Shark — Hagfishes, sharks and rays are marine vertebrate osmoconformers.

22. Osmoconformers are usually found in:

(a) Deserts (b) Mountains
(c) Rivers only (d) Marine water

Answer: (d) Marine water — Marine water has a relatively stable solute concentration.

23. Most vertebrates in aquatic and terrestrial environments are:

(a) Osmoconformers (b) Ammonotelic
(c) Osmoregulators (d) Poikilotherms

Answer: (c) Osmoregulators — They can regulate water and solutes in their body fluids.

24. Osmosis always occurs from:

(a) Hypertonic to hypotonic solution (b) Hypotonic to hypertonic solution
(c) Isotonic to isotonic solution (d) Concentrated to dilute solution

Answer: (b) Hypotonic to hypertonic solution — Water moves from the more dilute to the less dilute solution.

25. Freshwater animals are generally _____ to their environment.

(a) Hypertonic (b) Isotonic
(c) Hypotonic (d) Impermeable

Answer: (a) Hypertonic — Their body fluids have a higher solute concentration than the water.

26. Freshwater fishes produce:

(a) Small volume of concentrated urine (b) Uric acid crystals
(c) No urine (d) Large volume of dilute urine

Answer: (d) Large volume of dilute urine — They remove the excess water that enters by osmosis.

27. Freshwater animals make up the loss of salts by:

(a) Drinking seawater
(b) Active transport of salts into the body
(c) Producing concentrated urine
(d) Rectal glands

Answer: (b) Active transport of salts into the body — They actively transport salts back into their bodies.

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28. Most marine animals are _____ to their environment.

(a) Impermeable (b) Hypertonic
(c) Isotonic (d) Hypotonic

Answer: (d) Hypotonic — Seawater has a much higher salt concentration than their tissues.

29. In marine fishes, the excess salts are actively pumped out of the blood by:

(a) Chloride cells of gills (b) Sweat glands
(c) Oil glands (d) Liver cells

Answer: (a) Chloride cells of gills — Chloride cells use ATP to pump salts out.

30. Rectal glands of some marine fishes remove:

(a) Water (b) Urea
(c) Salts (d) Ammonia

Answer: (c) Salts — They pass salts from blood into the digestive tract.

31. Marine fishes conserve water by producing:

(a) Very small amount of highly concentrated urine
(b) Large volume of dilute urine
(c) Sweat
(d) Metabolic water only

Answer: (a) Very small amount of highly concentrated urine — This reduces the loss of water.

32. The waxy exoskeleton of insects is made of:

(a) Keratin (b) Urea
(c) Cellulose (d) Chitin

Answer: (d) Chitin — It prevents evaporative water loss.

33. The waterproof scales of reptiles are made of:

(a) Chitin (b) Keratin
(c) Wax (d) Calcium

Answer: (b) Keratin — Keratin scales reduce water loss.

34. Water produced as a by-product of cellular respiration is called:

(a) Osmotic water (b) Excretory water
(c) Metabolic water (d) Tissue water

Answer: (c) Metabolic water — The kangaroo rat survives entirely on it.

35. Which animal survives entirely on metabolic water?

(a) Kangaroo rat (b) Shark
(c) Tardigrade (d) Hagfish

Answer: (a) Kangaroo rat — It feeds on carbohydrate-rich seeds of desert plants.

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36. The dormant state of tardigrades after losing almost all body water is called:

(a) Hibernation (b) Anhydrobiosis
(c) Osmoconformation (d) Hypothermia

Answer: (b) Anhydrobiosis — They become active again when water returns.

37. Many desert animals stay underground during the day to avoid:

(a) Predators (b) Cold
(c) Rain (d) Evaporation

Answer: (d) Evaporation — They come out at night when the air is cooler and more humid.

38. One gram of ammonia requires how much water to dilute it to a non-toxic level?

(a) 1 mL (b) 50 mL
(c) 500 mL (d) 1000 mL

Answer: (c) 500 mL — Ammonia is highly toxic and needs plenty of water.

39. One gram of urea requires how much water for its excretion?

(a) 50 mL (b) 1 mL
(c) 500 mL (d) 5 mL

Answer: (a) 50 mL — Urea is less toxic than ammonia.

40. One gram of uric acid requires how much water for its excretion?

(a) 50 mL (b) 1 mL
(c) 500 mL (d) 100 mL

Answer: (b) 1 mL — Uric acid is the least toxic and least soluble.

41. Animals which excrete ammonia as their major nitrogenous waste are called:

(a) Ureotelic (b) Uricotelic
(c) Ammonotelic (d) Osmoconformers

Answer: (c) Ammonotelic — e.g., most fishes, protozoans, sponges, coelenterates.

42. Which of these animals is uricotelic?

(a) Sponge (b) Human
(c) Freshwater fish (d) Bird

Answer: (d) Bird — Insects, desert animals, birds and most reptiles excrete uric acid.

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43. The chemical formula of urea is:

(a) $CO(NH_2)_2$
(b) NH_3
(c) $C_5H_4N_4O_3$
(d) NH_4OH

Answer: (a) $CO(NH_2)_2$ — Urea is $CO(NH_2)_2$.

44. The chemical formula of uric acid is:

(a) NH_3
(b) $CO(NH_2)_2$
(c) $C_5H_4N_4O_3$
(d) $C_6H_{12}O_6$

Answer: (c) $C_5H_4N_4O_3$ — Uric acid is $C_5H_4N_4O_3$.

45. Ammonia is produced by the removal of which group from amino acids?

(a) Carboxyl group
(b) Amino group
(c) Hydroxyl group
(d) Phosphate group

Answer: (b) Amino group — The $-NH_2$ group combines with H^+ to form NH_3.

46. The conversion of ammonia into uric acid:

(a) Releases energy
(b) Occurs in gills
(c) Needs plenty of water
(d) Uses a lot of energy

Answer: (d) Uses a lot of energy — The energy cost is repaid by saving water.

47. In humans, uric acid is a by-product of the breakdown of:

(a) Glucose
(b) Fatty acids
(c) Purines
(d) Chitin

Answer: (c) Purines — Purines are found in many foods, particularly meat.

48. Deposition of uric acid crystals in joints causes:

(a) Goosebumps
(b) Gout
(c) Hyperthermia
(d) Inflammation of gills

Answer: (b) Gout — Excess uric acid in blood forms crystals in joints and tissues.

13.3 THERMOREGULATION

49. If temperature rises well above the optimal, enzymes:

(a) Work faster permanently
(b) Remain unchanged
(c) Become rigid and brittle
(d) Denature and lose their function permanently

Answer: (d) Denature and lose their function permanently — High temperature denatures enzymes.

50. Low temperature makes cell membranes:

(a) More permeable (b) Denatured
(c) Rigid and brittle (d) Thicker

Answer: (c) Rigid and brittle — The lipid bilayer is temperature-sensitive.

51. At high temperatures, cell membranes can lose their:

(a) Lipids (b) Selective permeability
(c) Proteins (d) Colour

Answer: (b) Selective permeability — This disturbs the working of cells.

52. Animals that gain most of their body heat from external sources are:

(a) Ectotherms (b) Endotherms
(c) Homeotherms (d) Osmoregulators

Answer: (a) Ectotherms — e.g., most invertebrates, fishes, amphibians and reptiles.

53. Which of these is an endotherm?

(a) Frog (b) Lizard
(c) Sparrow (d) Fish

Answer: (c) Sparrow — Birds and mammals are endotherms.

54. An ectotherm sitting on a warm rock gains heat by:

(a) Radiation (b) Conduction
(c) Evaporation (d) Metabolism

Answer: (b) Conduction — Basking in the sun is radiation; a warm rock gives heat by conduction.

55. Ectotherms regulate their body temperature mainly by:

(a) Behavioural thermoregulation (b) Shivering
(c) Burning brown fat (d) Sweating

Answer: (a) Behavioural thermoregulation — They bask, seek shade, burrow or become dormant.

56. In cold, endotherms may burn which tissue for heat?

(a) Epidermis (b) Bone tissue
(c) Nervous tissue (d) Brown adipose tissue

Answer: (d) Brown adipose tissue — They also shiver to produce heat.

57. Hair, feathers and fat help endotherms in:

(a) Heat production (b) Insulation (retaining heat)
(c) Sweating (d) Excretion

Answer: (b) Insulation (retaining heat) — Insulation reduces heat loss.

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58. Animals unable to maintain their body temperature within narrow limits are:

(a) Homeotherms (b) Endotherms
(c) Poikilotherms (d) Uricotelics

Answer: (c) Poikilotherms — Their temperature matches the environment.

59. All invertebrates are:

(a) Homeotherms (b) Endotherms
(c) Osmoregulators (d) Poikilotherms

Answer: (d) Poikilotherms — Fishes, amphibians and reptiles are also poikilotherms.

60. Humans are:

(a) Endothermic homeotherms (b) Endothermic poikilotherms
(c) Ectothermic homeotherms (d) Ectothermic poikilotherms

Answer: (a) Endothermic homeotherms — They make their own heat and keep it constant at about 37 °C.

61. The "thermostat" of homeotherms is the:

(a) Thyroid gland (b) Skin
(c) Hypothalamus (d) Pituitary gland

Answer: (c) Hypothalamus — It triggers cooling or heating responses.

62. Poikilotherms can survive in nutrient-poor environments because they:

(a) Do not need massive food to fuel a constant temperature
(b) Eat no food
(c) Produce uric acid
(d) Have brown fat

Answer: (a) Do not need massive food to fuel a constant temperature — A homeotherm would starve there.

63. Expansion of blood capillaries beneath the epidermis is called:

(a) Vasoconstriction (b) Vasodilation
(c) Shivering (d) Hypothermia

Answer: (b) Vasodilation — It occurs in overheating to remove more heat.

64. A rise of body temperature above 37 °C is called:

(a) Hypothermia (b) Anhydrobiosis
(c) Homeostasis (d) Hyperthermia

Answer: (d) Hyperthermia — A fall below 37 °C is hypothermia.

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65. Sweating cools the body because:

(a) Sweat is cold
(b) Evaporation of sweat carries heat from the body
(c) Sweat blocks the capillaries
(d) Sweat reduces thyroxine

Answer: (b) Evaporation of sweat carries heat from the body — Evaporation produces a cooling effect.

66. In overheating, the thyroid gland may decrease the secretion of:

(a) Thyroxine (b) Insulin
(c) Oxytocin (d) Cytokines

Answer: (a) Thyroxine — This lowers the basal metabolic rate and heat production.

67. Spasmodic contraction of skeletal muscles that produces heat is:

(a) Vasodilation (b) Sweating
(c) Shivering (d) Goosebumps

Answer: (c) Shivering — Shivering raises the body temperature.

68. Goosebumps help in thermoregulation because the erected hair:

(a) Produce heat (b) Widen capillaries
(c) Release sweat (d) Trap air which acts as an insulator

Answer: (d) Trap air which acts as an insulator — Trapped air reduces heat loss.

69. Vasoconstriction results in:

(a) Less heat loss from blood (b) More heat loss from blood
(c) More sweating (d) Lower metabolic rate

Answer: (a) Less heat loss from blood — Blood flow to the skin is reduced.

70. Humans regulate their body temperature through:

(a) Positive feedback (b) Behaviour only
(c) Osmosis (d) Negative feedback

Answer: (d) Negative feedback — The response reverses the change in temperature.

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